RAKSHIT KAPOOR

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EDUCATION

Vellore Institute of Technology, Vellore, India — Expected July 2027
Bachelor of Technology in Computer Science (Specialization in Information Security) **CGPA: 7.6 / 10.0**

TECHNICAL SKILLS

- **Programming Languages:** Java, C++, Python, SQL, JavaScript (ES6+)
- **Web Development & Frameworks:** Node.js, Express.js, ReactJS, HTML5, CSS3, RESTful APIs
- **Databases & Cloud:** MySQL, MongoDB Atlas, Cloud Fundamentals
- **Tools & Platforms:** Git, GitHub, Visual Studio Code, Wireshark, PyTorch, OpenCV
- **Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, DBMS, Information Security & Cryptography Principles

PROFESSIONAL EXPERIENCE

Full-Stack Developer Intern | Impinge Solutions | May – July
- Developed full-stack modules for a customer inquiry tracking and lead management platform designed for offline businesses.
- Architected RESTful APIs and database schemas using JavaScript (ES6+), Node.js, Express.js, HTML5, CSS3, and MySQL.
- Collaborated on version control workflows via Git/GitHub, ensuring clean code structure and modular API integration.

Conversational AI Intern | VOIS for Tech University Engagement Program
- Engineered conversational data analysis pipelines utilizing Large Language Models (LLMs) to analyze text interactions and extract insights.

AI & Cloud Technology Intern | AICTE Virtual Internship Program
- Completed hands-on learning activities and industry modules covering artificial intelligence fundamentals, machine learning workflows, and cloud architecture.

TECHNICAL INNOVATION PROJECTS

AI-Powered Diabetic Retinopathy Screening and Diagnosis System (RetinaSight) | Conference paper applied
- Co-authored a conference paper on an end-to-end deep learning system (RetinaSight) for safety-aware automated diabetic retinopathy screening.
- Engineered a paired regression-classification head on an EfficientNet-B3 backbone, achieving a Quadratic Weighted Kappa (QWK) score of 0.9334 and 98.64% referable-disease sensitivity on the APTOS 2019 benchmark.
- Integrated dual safety gates (retinal validity gate using ResNet-18 and photometric heuristics, plus image quality assessment for blur/exposure) to prevent misgrading out-of-distribution or ungradable inputs.
- Incorporated Grad-CAM visual attention overlays, per-class reliability reporting, and a 3-tier RESTful web architecture (Node.js/Express, Python/FastAPI, MongoDB) with PDF clinical report generation.

Anonymous Help Request & Incident Platform | Node.js, Express.js, REST APIs, Web Security
- Architected a full-stack secure platform for anonymous incident reporting and request submission.
- Designed and implemented REST APIs to support end-to-end request lifecycle management and real-time status tracking.

Currency Note Detection Web Interface | HTML5, CSS3, JavaScript, API Integration, Computer Vision
- Built a responsive web interface enabling seamless image uploads and integrated classification APIs for real-time currency verification.

Lecture Voice-to-Notes Generator | Speech Recognition APIs, Web Audio, JavaScript
- Conceptualized and developed an automated application to convert complex lecture audio into structured, concise Markdown notes.

CERTIFICATIONS, HACKATHONS & LEADERSHIP

- **IBM Artificial Intelligence Fundamentals Certification:** IBM — Demonstrated core mastery in AI concepts, machine learning models, and ethical AI deployment.
- **Hackathons & Open-Source Contributions:** Active participant in collaborative technical hackathons and developer communities; contributor to open-source software and tools on GitHub (RakK-42005).
- **Demonstrated Technical Leadership & Problem-Solving:** Led cross-functional academic project teams, establishing system architectures, safety gates, and technical documentation standards.